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Experimental and Numerical Investigations on Fabrication of Surface Microstructures Using Mask Electrolyte Jet Machining and Duckbill Nozzle

Ming Wu
Department of Mechanical Engineering;
Department of Computer Science,
KU Leuven, 3001 Leuven, Belgium
e-mail: ming.wu1@kuleuven.be

Muhammad Hazak Arshad
Department of Mechanical Engineering,
KU Leuven, 3001 Leuven, Belgium
e-mail: muhammadhazak.arshad@kuleuven.be

Krishna Kumar Saxena
Department of Mechanical Engineering,
KU Leuven, 3001 Leuven, Belgium
e-mail: krishna.saxena@kuleuven.be

Dominiek Reynaerts
Department of Mechanical Engineering,
KU Leuven, 3001 Leuven, Belgium
e-mail: dominiek.reynaerts@kuleuven.be

Zhongning Guo
College of Mechanical and Electrical Engineering,
Guangdong University of Technology,
Guangzhou 510006, China
e-mail: znguogdut@163.com

Jiangwen Liu[1]
College of Mechanical and Electrical Engineering,
Guangdong University of Technology,
Guangzhou 510006, China
e-mail: fejwliu@scut.edu.cn

The consistency in the fabrication of microsurface structures on large workpieces remains a challenge for existing production techniques. Mask electrolyte jet machining (MEJM) is a hybrid mask-based electrochemical machining (ECM) process that combines the flexibility of Jet-ECM to flush the electrochemical by-products and the high throughput processing feature of through-mask electrochemical micromachining (TMEMM). In the present study, a duckbill-shaped nozzle is employed in the MEJM for the batch fabrication of microsurface structures which facilitates more uniform current density distribution over the entire machining area, resulting in better consistency. With a larger slit length, the duckbill nozzle will not only cover more processing area but also facilitates more uniform current density distribution over the entire machining area, resulting in a better consistency for batch fabrication. Thresholds of the ratio of slit length to the machining area were derived from a quantitative analysis, namely the efficient threshold and the performance threshold. The slit length of the duckbill nozzle should be at least twice as large as the machining area to wipe out any observable deviation on current density distribution in the machining area. An efficient and high-performance numerical simulation framework with a virtual gap concept is developed for the mask-based ECM processes to simulate microcavity profiles and associated current density distribution. The concept of a virtual gap is proposed to solve the simulation dilemma of elements being consumed in the mask-based ECM process. Quantitative analyses were carried out to study how the virtual gap influences the electric current density distribution in the interelectrode gap. A virtual gap smaller than 100 nm is recommended. Guidelines on how to ensure a smooth electric field transition across the coarse and fine-meshed zones are presented by conducting a quantitative analysis. As an example, this work has successfully fabricated several cavity arrays with different processing parameters. Both the experimental results and the numerical simulation frameworks are easy-to-implement and easy-to-extend for all the mask-based ECM processes. [DOI: 10.1115/1.4056570]

Keywords: electrical and electrochemical machining, machining processes, micro- and nano-machining and processing, modeling and simulation, nontraditional manufacturing processes

1 Introduction

Electrochemical machining (ECM) is a promising technique for manufacturing aerospace components [1], household appliances [2], MEMs products [3], and other industrial products [4,5]. A wide range of conductive materials, regardless of the hardness, such as dual-phase steel [6], high entropy alloys [7], titanium alloys [8], and carbide alloys [9] can be machined effectively by ECM as it is free from tool wear and does not generate any heat-affected layers or residual stresses on the workpiece surface. In ECM, only the metallic materials are dissolved, it is widely used to produce accurate surface features with nonconductive masks having the desired pattern applied on the workpiece surface by leveraging the photolithography technology. Through-mask electrochemical micromachining (TMEMM) is a representative ECM-lithography hybrid machining method, for instance, Qu et al. [10] presented a study on microsurface fabrication on an inner cylindrical surface by TMEMM. Singh Patel et al. [11] presented a porous and flexible tool in TMEMM to produce microsurface structures on the free-form surfaces. In ECM, the electrolytic products, which have negative effects on the electrochemical reactions, will accumulate as the process progress [12] and need to be carried away in time with the electrolyte flow. In that respect, TMEMM naturally suffers from a poor flushing condition in the machining area because of its workpiece holding system—the electrolyte flow direction is normal to the mask, which would lead to a very low flowrate in the mask holes [13].

In ECM, the process machining accuracy and surface quality are highly related to the electrolyte flow condition [14], studies to improve the electrolyte flow field in TMEMM for processing consistency are carried out intensively in the last few years. In 2017, Wang et al. [15] found that the electrolyte flow speed would be faster at the bottom of machining when using a mask with cone-shaped holes instead of classical cylindrical throughput holes. In 2019, Li et al. [16] proposed a holding system to force the electrolyte flow along a zig-zag path to increase the machining consistency. However, despite the sophisticated and well-designed holding system, the one-way electrolyte flow field is still there and can yield "flow marks" [17] which are in many cases undesirable. In 2022, Liu et al. [18] proposed a vibrated tool together with

[1]Corresponding author.

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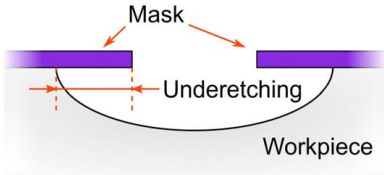

Fig. 1 Underetching in the mask-based ECM process

additional electrolyte inlets to increase the electrolyte flow conditions in the interelectrode gap (IEG). These inlets were distributed in different directions, the electrolyte flow direction can be changed, significant "flow marks" reductions were observed. Inspired by Jet-ECM [19,20], Wu et al. [21] proposed a method named mask electrolyte jet machining (MEJM) which is using a moving nozzle as the cathode. MEJM is providing a more flexible and affordable way to optimize the electrolyte flow within the mask space.

Despite the applications that have been realized and detailed fundamental studies [22,23], fetching the numerical simulation results in the mask-based ECM process, two major hurdles remain to be solved in order to be able to do an in-depth computational analysis on the mask-based ECM process:

(i) The mask-based ECM methods, including TMEMM and MEJM, are multiscale processes. The features on the photoresist mask are in the order of microns, while the overall area to be processed, i.e., the overall size of microsurface structures is in the order of millimeters. In addition, the accuracy of numerical results is directly tied to the mesh size, hence, it will be a computationally intensive work when connecting different domains with different dimension scales. In addition, with the advancing lithography, the batch fabrication of submicrosurface structures by the mask-based ECM process has been realized in experimental investigations [24,25]. A proper method that can perform multiscale numerical simulations with fair computational costs becomes an emerging need for further industrial applications.

(ii) As the machining process proceeds, the electrochemical reaction interface is "stripped off" from the mask. Therefore, the generation of open space between the mask and workpiece is inevitable. The material removal process on the reaction interface usually is described by the arbitrary Lagrangian–Eulerian (ALE) [16,26] method, where the "elements" can be deformed but cannot be created or consumed during the simulation process. This dilemma makes it impossible to simulate the underetching as shown in Fig. 1. In Ref. [27], the island effect in TMEMM was well studied by a multiphysical model but could not help to illustrate the underetching. In Ref. [28], a re-useable PDMS mask was employed in TMEMM. The evolution of electric current density distribution on the workpiece surface was well-studied, but the underetching was yet absent.

In the present work, a duckbill-shaped nozzle is employed in the MEJM to perform a batch fabrication of microsurface structures. In order to present an efficient and high-performance simulation, a discretized model with a virtual gap presumption is developed to reduce the computational costs and ensure a proper boundary motion to describe the material removal process under the mask. The physical model is subdivided into different pieces with independent mesh sizes, depending on the need: a finer mesh for the domains that we are interested in, and a coarser mesh for the domains that are less essential. This kind of approach has been a common strategy for solving multiscale problems [29,30]. Quantitative analyses were carried out to provide guidelines for the mesh size selection to ensure an effective electric field transition across the mesh mismatching interface. The underetching in the mask-based ECM process can be performed in a high resolution.

Quantitative analyses were carried out to inspect how the virtual gap influences the electric current density distribution in the interelectrode gap. The equivalent physical meaning for the different sizes of the virtual gap was also studied. A virtual gap smaller than 100 nm was recommended.

Along with the theoretical analysis, experimental studies were also carried out. This duckbill nozzle is initially proposed to improve the processing efficiency by covering more processing area but, it has been found that the processing consistency is improved as well. The relationship between the ratio of the duckbill nozzle size to the machining area and processing consistency is investigated by a quantitative analysis. The capability of simulating the underetching by the present model was verified by experiments as well.

2 Process Principle and Theoretical Model

2.1 Process Principle and Experimental Setups. Figure 2 shows the process schematic of MEJM with a duckbill nozzle. During the machining process, the high-speed electrolyte jet (sodium nitrate solution in this case) flows through the hollow metallic duckbill nozzle and the workpiece with the required mask pattern (in this case several microdimple arrays were developed by lithography with the Shipley 1818 photoresist, the detailed

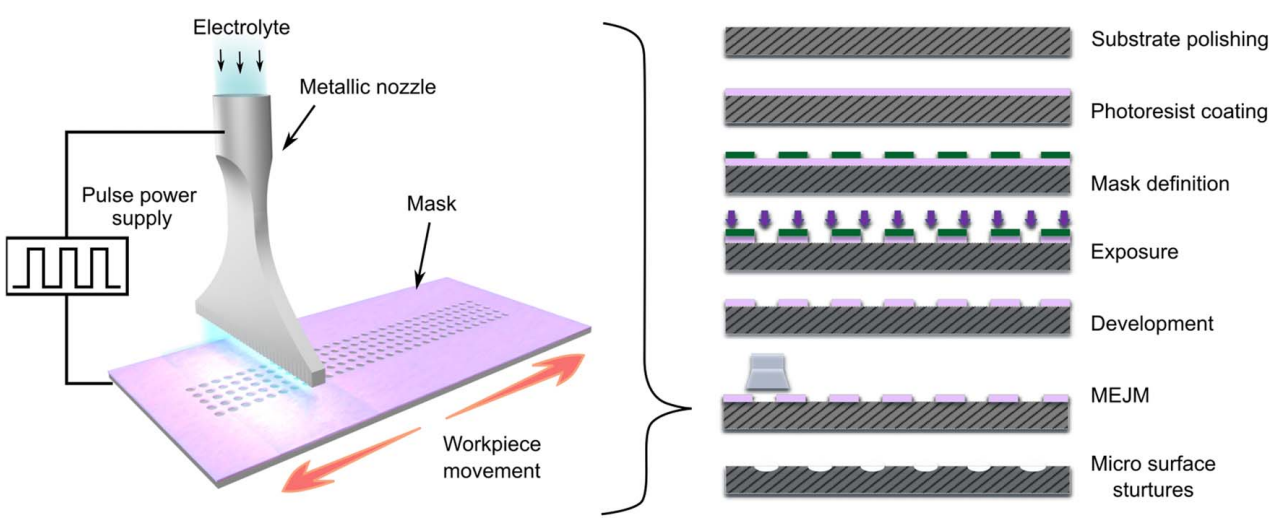

Fig. 2 Schematic view of the MEJM process

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lithography process is provided in the Appendix) moves under it in the desired direction. The workpiece surface under the microholes of the mask is dissolved when the high-speed electrolyte jet contacts the exposed workpiece surface upon application of sufficient voltage. As a result, the microdimple array can be generated according to the mask pattern by moving the workpiece under the metallic duckbill nozzle. The specific advantage of a metallic nozzle is that it allows to provide a stable and high-speed electrolyte jet flow for the workpiece as well as the renewal of electrolyte in the small machining area, which is useful for generating microcavities. The width of the metallic nozzle is kept considerably larger than the width of the mask pattern (for details, see Sec. 3.1) to ensure constant current density across all the microdimples for uniform dissolution. This method is highly flexible and enables large area machining.

As shown in Fig. 3, the major peripherals of the MEJM system used in this experiment include a control system (Leadshine Tech Co. 2410C motion control card), an electrolyte circulation filtering system, and a machine-tool frame (assembled by Beijin Padiwei Instrument Co.), which can realize linear motion in three directions of X, Y, and Z.

2.2 Model Description. For ECM, the electrochemical dissolution and consequently the machining accuracy is sensitive to the electric field and electrolyte flow condition. Compared to TMEMM, both the electric field and the electrolyte flow field in MEJM are time-dependent. Therefore, a model which can investigate the evolution of the current density distribution coupled with the electrolyte flow in the time domain is proposed.

The following three simplifications and presumptions are made:

(1) In the MEJM process, several arrays are processed at the same time. For the sake of simplicity, our analysis considers only the shape evolution of one dimple.
(2) Since both the workpiece and the duckbill nozzle are symmetric in the moving direction, the physical model can be reduced from 3D to 2D in order to save computational time.
(3) For MEJM, a constant voltage is applied in the simulation so the time-dependent effects due to the pulsed voltage are not considered. The potential distribution can be described using Laplace's equation

$$\nabla^2\phi = \frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\phi}{\partial y^2} = 0 \tag{1}$$

where ϕ is the electric potential and the x- and y-directions are the horizontal and vertical spatial axes, respectively, as shown in Fig. 4.

2.2.1 Discrete Modeling. In the present work, the diameter of the holes on the through-mask surface is less than 10^2 μm, while the nozzle travels around 10^4 μm on the workpiece surface. The accuracy of numerical analysis is directly related to the

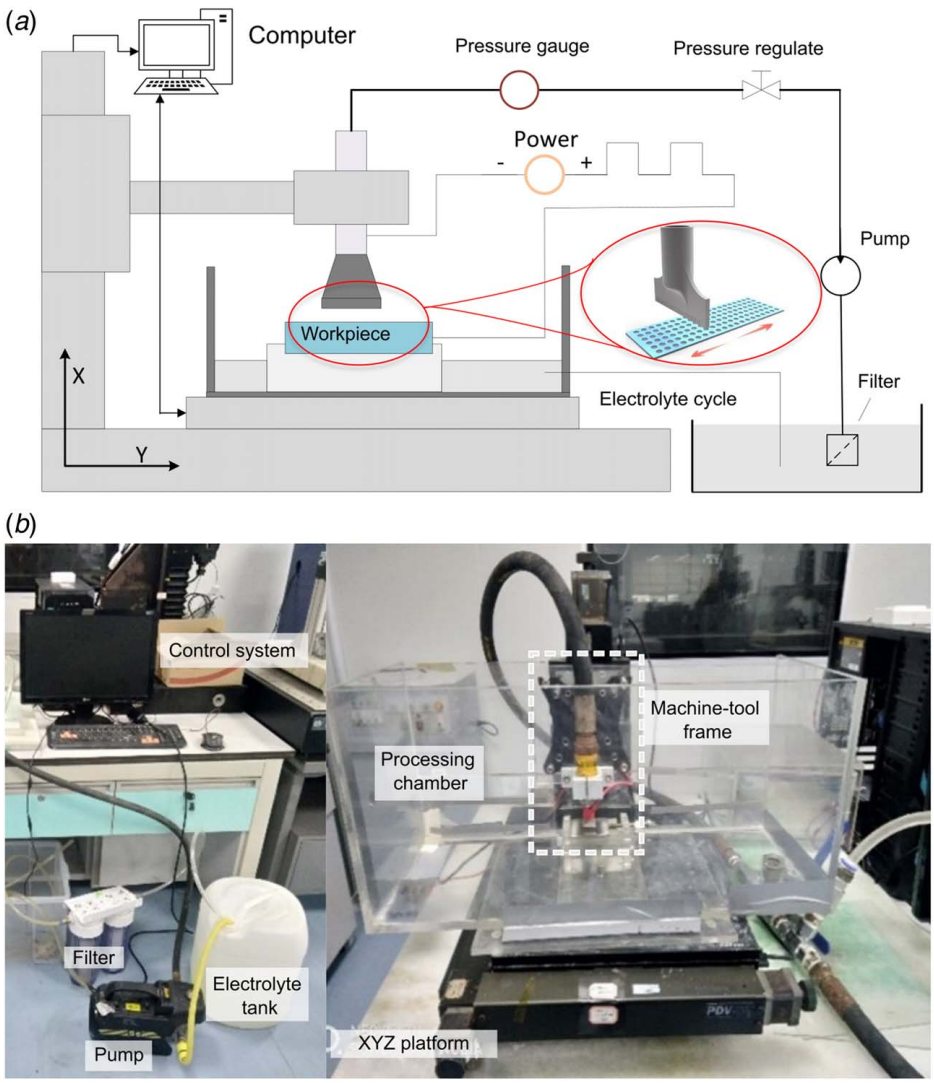

Fig. 3 (*a*) MEJM system with major peripherals and (*b*) experimental setup

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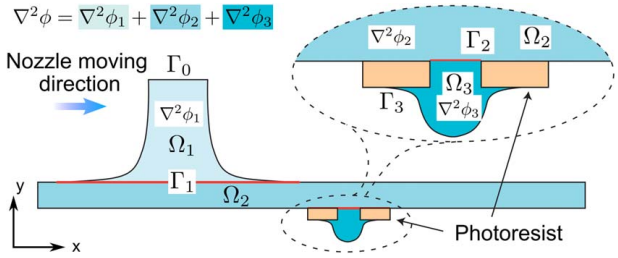

Fig. 4 Electric potential distribution in MEJM with subdomains

discretization's adequateness. However, the computational effort will increase as the mesh becomes finer. For MEJM, connecting different length scales over two orders of magnitude, i.e., the micro-to-macro connections, make it computationally intensive work if using a single unified physical model. One approach to resolving this problem is to discretize the physical model and use a fine mesh only in the region of interest [31,32].

As the Laplace's equation is linear and homogeneous, when the physical model is subdivided into pieces, each of the pieces is a solution to the Laplace's equation. Therefore, the sum of the individual pieces will also be a solution. In the present study, the electrolyte domain was subdivided into three subdomains, namely: (a) electrolyte jet domain Ω_1; (b) standing water domain Ω_2; (c) workpiece deformation domain Ω_3, as illustrated in Fig. 4 and Eq. (2)

$$\begin{aligned}\nabla^2\phi &= \sum \nabla^2\phi_i \\ &= \nabla^2\phi_1 + \nabla^2\phi_2 + \nabla^2\phi_3\end{aligned} \tag{2}$$

where the Γ_0 and Γ_3 denote the boundary conditions for cathode and anode, respectively. Γ_2. denotes the interface between the standing water domain Ω_2 and the workpiece deformation domain Ω_3. The potential distribution of workpiece deformation domain $\nabla^2\phi_3$ also satisfies Laplace's equation, so Eq. (1) can be rewritten as

$$\begin{cases}\nabla^2\phi = \dfrac{\partial^2\phi}{\partial x^2} + \dfrac{\partial^2\phi}{\partial y^2} = 0 \\ \phi \mid \Gamma_0 = 0V \\ \phi \mid \Gamma_3 = 10V\end{cases} \tag{3}$$

For the workpiece deformation domain (Ω_3), the potential distribution ϕ_3 of the machined cavity can be solved from Eq. (2) when $\phi_3 \mid \Gamma_2$ or $\phi_2 \mid \Gamma_2$ is confirmed. In the same way, for the electrolyte jet domain ($\nabla^2\phi_1$) and standing water domain ($\nabla^2\phi_2$), the electric distribution can be determined when the shared boundary condition is defined. Therefore, the multiscale problem in MEJM can be solved by building an assembly model in separate independent parts with some shared boundaries. The mesh in the workpiece deformation domain Ω_3, which is of the highest significance, should be fine enough to describe electrochemical dissolution precisely. Whereas the mesh in the standing water domain Ω_2 can be coarse to save computational power.

However, this strategy could raise concerns on the fidelity of electric field transition between coarse and fine meshes. In practice, when using finite element methods to get a numerical solution, like the floating-point error [33], mesh mismatching across the interface is inevitable when assembling meshes with independently determined sizes.

A physical model, as illustrated in Fig. 5, is built to quantify the reliability of the assembly model with different mesh sizes. The quantification is done by determining the deviation of current density distribution on the interface Γ_2 between both sides. In this model, a classical mask-based ECM process is simulated. To be consistent with the aforementioned boundary conditions in Eq. (3) and Fig. 4, Γ_0, Γ_3, and Γ_2 represent the anode ($U \mid \Gamma_0 = 0\,\mathrm{V}$), cathode ($U \mid \Gamma_3 = 10\,\mathrm{V}$), and the interface between standing water

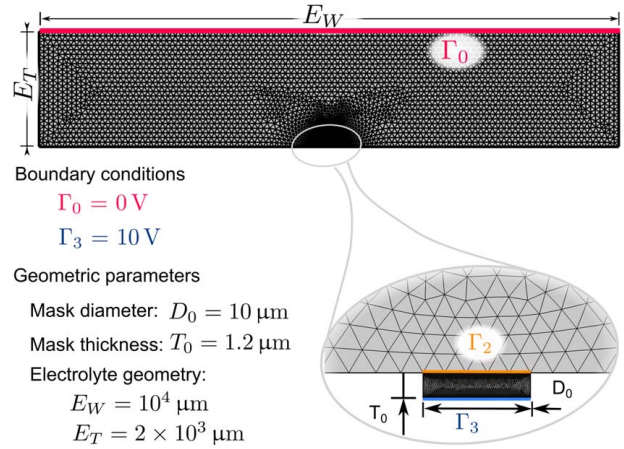

Fig. 5 Investigating model for simplification of mesh with continuity presumption

domain Ω_2 and workpiece deformation domain Ω_3, respectively. The mesh sizes across the interface Γ_2 are different: in the workpiece deformation domain Ω_3 side, the interface Γ_2 is divided into 100 pieces; whereas in the standing water domain Ω_2, the division number ranges from 1 to 100 with a step of 1, respectively.

The simulation results and corresponding $\mathbf{J}_i$ (division number = i) on Γ_2 with different division numbers are shown in Fig. 6. The relative deviation ratio $P_{\Delta J_i}$ of electric current density $\mathbf{J}$ is given by Eq. (4) and is shown in Fig. 7

$$P_{\Delta J_i} = \frac{\boldsymbol{J}_0 - \boldsymbol{J}_i}{\boldsymbol{J}_0} \times 100\% \tag{4}$$

The interface Γ_2 on the standing water domain side Ω_2 is divided by a fixed number of 100, hence, the baseline for the deviation ratio calculation is at division number = 100, i.e., $P_{\Delta J_{100}}$.

As represented in Figs. 6 and 7, both the current density deviation and relative deviation ratio are decreasing as the division number increases. To be specific, at the central area of the through-hole which is enclosed by the photoresist, the current density J is lower than the baseline; and the peak of relative deviation ratio is up to about 4% when the division number is 1. Whereas in the edge area of throughput hole, the current density j is higher than the baseline, which decreases with the division number. In the meantime, the increasing or decreasing trend of j demonstrated a nonlinear relationship with the varying division number. As the division number increases from 1 to 20, the electric current density distribution fits the j_{100} distribution rapidly, then the fitting rate slows down and finally approaches the baseline. For mask ECM due to the island effect [27] with a relative deviation ratio of about 4%, the calculation error for current density in the center and edge of the microcavity will lead to a qualitative deviation. Therefore, the errors which follow the mesh mismatching in the numerical simulation process should be as small as possible. In Figs. 6(*b*) and 7(*b*), a nonlinear color map is adopted due to this.

The relative deviation ratio $P_{\Delta J_i}$ depicts the deviation ratio for each coordinate point demonstrated as an array which is less intuitive to compare the performance with different division numbers, therefore, the deviation norm $|\mathbf{D}_i|$ is calculated as Eq. (5) and shown in Fig. 8

$$\begin{aligned}|\mathbf{D}_i| &= \sum_{k=1}^{n} |P_{\Delta J_{i_k}}| \\ &= \sum_{k=1}^{n} \left| \frac{\boldsymbol{J}_{0_k} - \boldsymbol{J}_{i_k}}{\boldsymbol{J}_{0_k}} \times 100\% \right|\end{aligned} \tag{5}$$

where k denotes the kth point on the interface Γ_2. As indicated in Fig. 8, when the division number increases from 1 to 5, the

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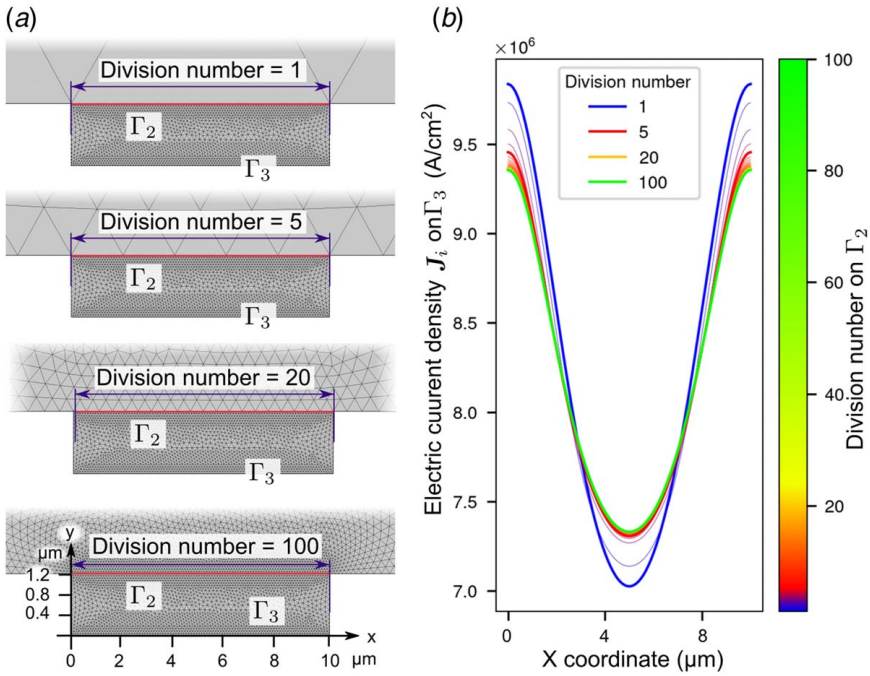

Fig. 6 Electric density distribution on Γ_2 with different division numbers: (*a*) mesh for numerical simulation and (*b*) electric current density with different division numbers on Γ_2

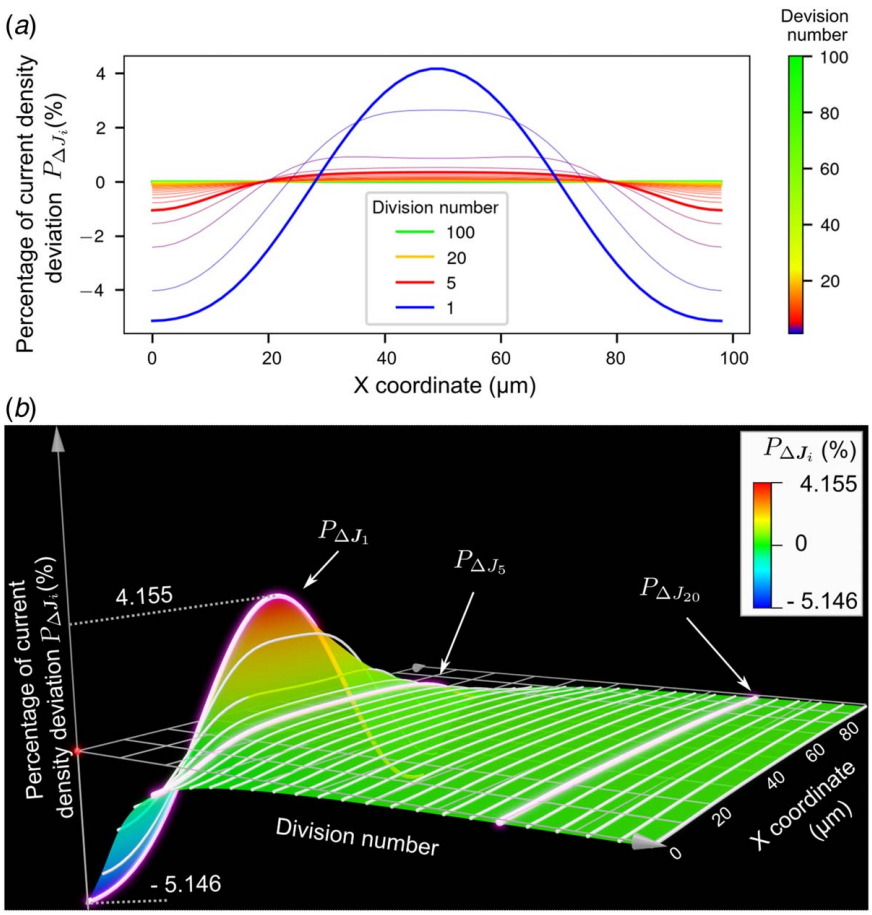

Fig. 7 Relative deviation ratio $P_{\Delta J_i}$ of electric current density on Γ_2 with different division numbers. The baseline for ratio calculation is the current density at division number = 100: (*a*) plot of relative deviation ratio $P_{\Delta J_i}$ of electric current density at different division numbers and (*b*) 3D view of relative deviation ratio $P_{\Delta J_i}$ of electric current density at different division numbers along with *X* co-ordinate

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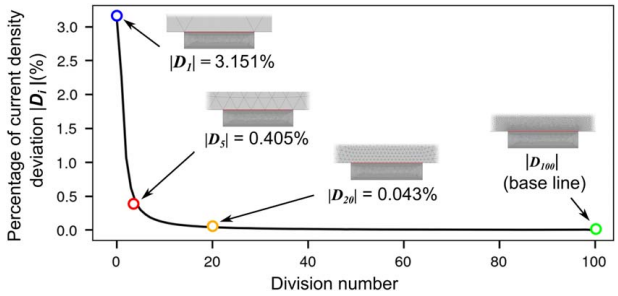

Fig. 8 Deviation ratio norm $|\mathbf{D}_i|$ of electric current density on Γ_2 with different division numbers. The baseline for ratio calculation is the deviation ratio norm at division number = 100.

deviation norm $|\mathbf{D}_i|$ drops rapidly (from 3.151% to 0.405%), but then flattens at division number greater than 20 ($|\mathbf{D}_{20}| = 0.043\%$).

All the results demonstrated that the adoption of a division number higher than 20 can ensure a good electric field transition through the nonmatching meshes. On the fine–coarse interface, the amount of nodes on the coarse mesh side should at least be 20% of the amount of nodes on the fine mesh side. In the present study, a division number of 40 is adopted for further investigations.

2.2.2 Virtual Gap in Mask-Based Electrochemical Machining. The deformation of the workpiece domain and consequently the moving boundary is usually solved by the ALE finite element method, where the "elements" can be deformed but, in principle, cannot be generated or eliminated during the simulation process. However, to illustrate the workpiece material removal process which takes place between the workpiece and photoresist mask, there must be some interface where the workpiece and photoresist mask can deform in their respective directions, thus the generation of an empty space between them is inevitable.

In order to solve this paradox, a virtual gap is presented as shown in Fig. 9 to ensure that the proper boundary movement corresponds to the principle of the ALE method. A physical model is built to quantify the reliability of a virtual gap with the geometry, boundary conditions and material properties identical to the ones adopted in Fig. 5 with the addition of a virtual gap to the model.

Since MEJM as a mask-based ECM variant, the material removal process is driven by electrochemical reactions and can be described with Faraday's law

$$\Delta m = \eta \frac{M}{zF} Q \tag{6}$$

where Δm is the weight change of workpiece, M is the molar mass of the workpiece material, z is the valency, F is Faraday's constant, and Q is the electric charge.

In the present study, all the elements in Eq. (6) except Δm and Q are identical for different virtual gap distances. Hence, as depicted

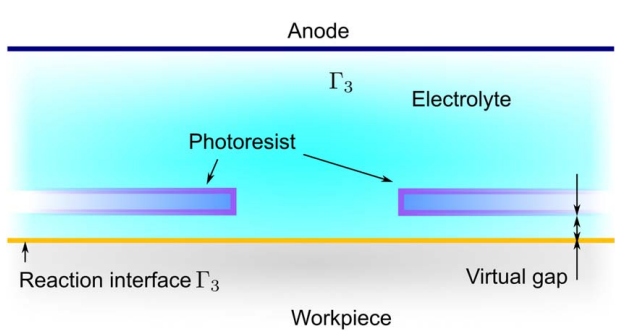

Fig. 9 Investigating model for simplification of mesh with continuity presumption

in Eq. (7), the material removal rate MRR by which the final profile is resolved can be simplified as a proportion to the electric current density $\boldsymbol{J}$ on the workpiece deformation interface Γ_3. Therefore, the influence of a virtual gap on the simulation profiles can be quantified by inspecting $\boldsymbol{J} \mid \Gamma_3$

$$\begin{aligned} \Delta m &= \eta \frac{M}{zF} Q \\ &= \eta \frac{M}{zF} \iint \boldsymbol{J}\, dA\, dt \\ \Rightarrow MRR &\propto \boldsymbol{J} \end{aligned} \tag{7}$$

The simulation results and $\boldsymbol{J}$ for different virtual gap distances are shown in Fig. 10. The virtual gap distances were varied from 5 nm to 1000 nm with a step of 5 nm. As indicated in Fig. 10, there is a qualitative change in the current density distribution. When the virtual gap is small (< 500 nm), the electric current density $\boldsymbol{J}$ at the edge of the through-hole enclosed by the photoresist is high, with two spikes and $\boldsymbol{J}$ at the central area is low. However, as the virtual gap grows over 500 nm, the peak current density appears only in the central area.

The current density is related to the geometry, the sharper the edge, the higher the current density. Hence, the current density spikes fade and are leveled off as the virtual gap distance increases. As shown in Figs. 10(*b*) and 11, there is little difference between $\boldsymbol{J}_5$ and $\boldsymbol{J}_{100}$, the difference is mainly visible by a reduction of the height of the spikes. Although it is true that these fading spikes are a representation of errors, the possibility that this is also a real association with the actual processing cannot be ruled out. For practical reasons (electrolyte flushing, weak adhesion between photoresist and metallic surface, etc.), there should be a small space at the edges of the through-hole mask where it is contacting the workpiece. Therefore, the electric current density spikes can only represent the case that the photoresist mask and the workpiece fit perfectly with no gap.

Here, the deviations between two current densities are presented as $\Delta\boldsymbol{J}$ and are shown in Fig. 11. However, even a graphical representation of these deviations in 3D (Fig. 11(*b*)) is not sufficiently intuitive to study the influences of different virtual gaps in the machining process. As the electric current density on some regions of Γ_3 is 0, therefore, the relative error for each coordinate point cannot be calculated by Eq. (4). The Euclidean distance $\|\Delta\boldsymbol{J}_i\|$ given in Eq. (8) is used to quantify the current density for better visualization of differences between the lines. The baseline for calculation is $\boldsymbol{J}_0$. In addition, the slope of $\|\Delta\boldsymbol{J}_i\|$ given by Eq. (9) was also investigated

$$\begin{aligned} \|\Delta\boldsymbol{J}_i\| &= d(\boldsymbol{J}_0, \boldsymbol{J}_i) \\ &= \sqrt{\sum_{k=1}^{n} |\boldsymbol{J}_{0_k} - \boldsymbol{J}_{i_k}|^2} \end{aligned} \tag{8}$$

$$\begin{aligned} m_{\|\Delta\boldsymbol{J}_i\|} &= \frac{\|\Delta\boldsymbol{J}_{i+1}\| - \|\Delta\boldsymbol{J}_i\|}{i+1-i} \\ &= \|\Delta\boldsymbol{J}_{i+1}\| - \|\Delta\boldsymbol{J}_i\| \end{aligned} \tag{9}$$

From the slope $m_{\|\Delta\boldsymbol{J}_i\|}$ depicted in Fig. 12(*b*), there is a clear transition in terms of slope denoted by a dashed line, which illustrates that the $\|\Delta\boldsymbol{J}_i\|$ can be categorized with high slope and low slope.

As indicated earlier, since the changes of current density distribution are only associated with the topological difference in this numerical simulation, hence, the changes of $\|\Delta\boldsymbol{J}_i\|$ are products of:

(i) the drastic change of geometric curvature at the edge of the through-hole;
(ii) the increase of interelectrode gap as a larger virtual gap.

In the meantime, the perfect fit between the photoresist and workpiece is either absent in practice or would be absent immediately as the electrochemical dissolution starts. Therefore, introducing a

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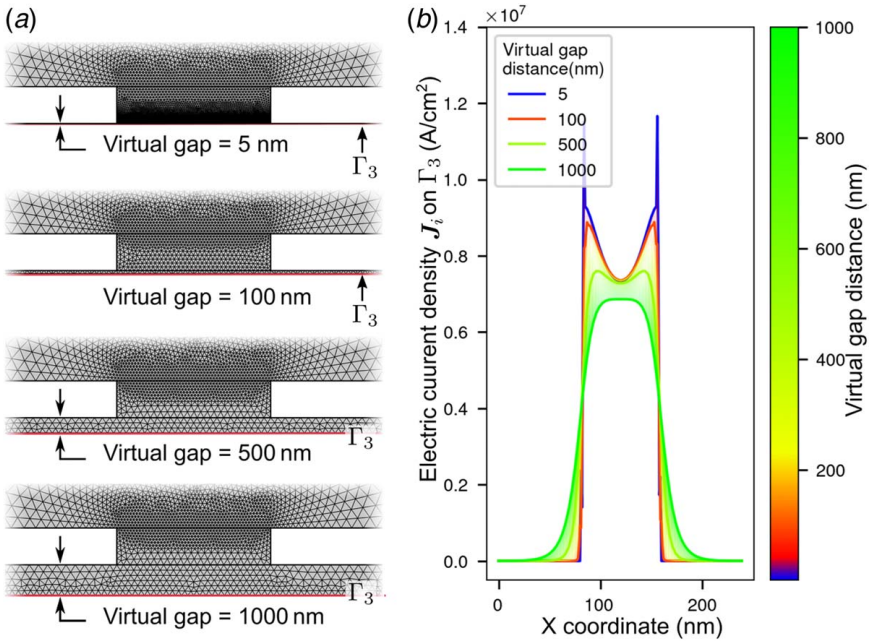

Fig. 10 Electric current distribution on Γ_3 with different virtual gap distances: (*a*) mesh for numerical simulation and (*b*) electric current density with different virtual gap distances on Γ_3

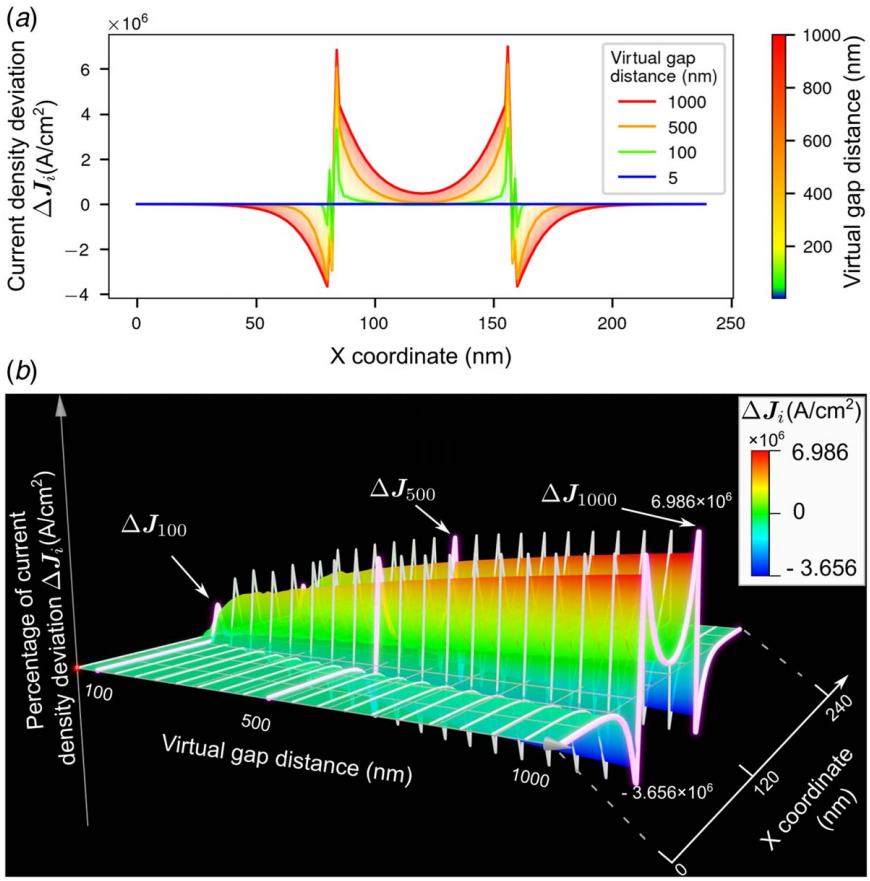

Fig. 11 Electric current density deviation ΔJ_i with different virtual gap distances. The baseline for calculation is the current density at virtual gap distance = 5 nm: (*a*) plot of electric current density deviation ΔJ_i with different virtual gap distances and (*b*) 3D view of percentage of electric current density deviation ΔJ_i with different virtual gap distances along with *X* co-ordinate.

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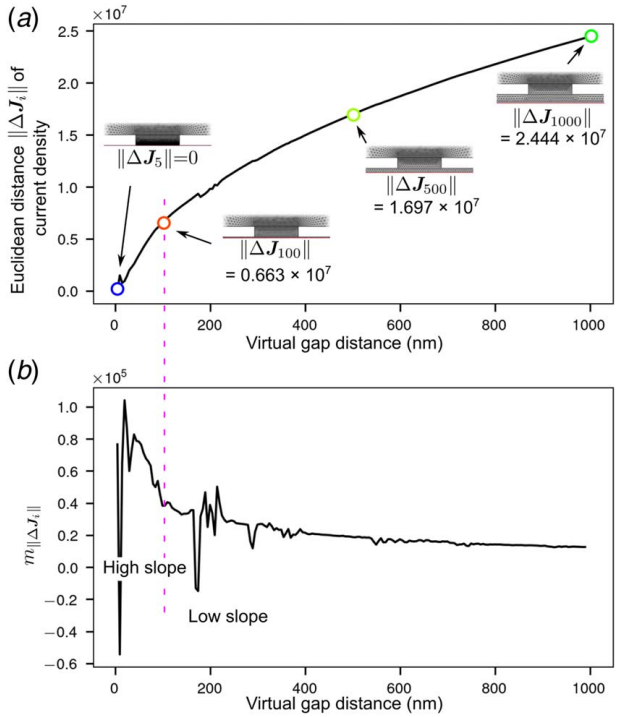

Fig. 12 Euclidean distances $||\Delta J_i||$ of electric current density and its slope. Dotted line: threshold for the acceptable virtual gap: (*a*) euclidean distance $||\Delta J_i||$ of current density on Γ_3 with different virtual gap distances and (*b*) slope of $||\Delta J_i||$.

virtual gap, i.e., smaller than 100 nm, which only eliminates the spike of the high slope domain in Fig. 12, might be justified.

In the present study, a virtual gap of 100 nm was employed, the deviation caused by altering the interelectrode gap is small, and the imperfections in the process are well assumed.

Based on the quantification of the errors, it can be concluded that the physical models using the aforementioned approach are allowing to perform a theoretical analysis with acceptable margins.

3 Numerical Simulation and Experimental Results

3.1 Investigation of the Duckbill Nozzle Effects on the Processing Consistency. The duckbill nozzle presented in this work facilitates a more efficient large-scale processing capability, which can uniformly machine additional processing areas.

Experimental investigations were conducted on 304 stainless steel workpiece to evaluate the ECM dissolution performance improvement using the duckbill nozzle. The workpiece was polished to have a mirror-like surface to enable a uniform contact with the mask. The lithographic mask array holes are 20 μm in diameter and distributed as several 35 × 35 square arrays with the distance between the hole centers as 150 μm, to fit in the working space of lithography technology of a 3-in diameter circle. A "duckbill-shaped" nozzle is what the jet nozzle adopts. There are two models of different dimensions: one had a slit width of 7 mm with a slit length of 11 cm. And the other had the same slit width but a slit length of 7 cm. The experimental parameters used in the experimental study are shown in Table 1.

The typical processing results are shown in Fig. 13. The surface topography of the microcavities was observed with a scanning electron microscope (S-3400N(II)). However, since the processing accuracy in MEJM is dominated by the accuracy of mask, microcavities machined by a duckbill nozzle of different sizes just look identical, the results shown in Fig. 14 could hardly reveal any significant information. Therefore, the machined dimensions of microcavities were measured by a laser scanning confocal microscope

Table 1 Processing parameters for investigations on effects of duckbill width

Parameters	Value
Voltage, U	20 V
Pulse frequency, f	1 kHz
Pulse duty cycle	50%
Nozzle moving speed, v	2 mm/s
Interelectrode gap	2 mm
Electrolyte concentration	10% (wt%) aq. $NaNO_3$
Electrolyte temperature	25 °C
Electrolyte pressure, P	20 kPa
Mask diameter, D_0	20 μm
Mask thickness, T_0	1.2 μm
Workpiece material	Stainless steel 304

(Olympus OLS-4100), 50 cavities were randomly selected for measurements.

The diameter D and depth h of microcavities were measured, as shown in Fig. 13. The aspect ratio AR and the etch factor EF, the ratio of material removal amount in depth and underetching, were calculated as shown in formulas (10) and (11). The corresponding standard deviations of 50 measurements for diameter, depth, and EF (Std_D, Std_h, and Std_{EF}) were also calculated to quantify the process consistency

$$AR = \frac{h}{D} \tag{10}$$

$$EF = \frac{2h}{D - D_0} \tag{11}$$

The experimental results are shown in Table 2 and kernel density estimation (KDE) plots as shown in Figs. 15 and 16 are presented for a much more intuitive and accurate understanding of the shape of the data distribution. The multivariate KDE-based probabilistic density function (PDF given in Eq. (12)) described in Ref. [34] is used to investigate the processing pattern [35] in MEJM. It evaluates the distribution of all the data collected and is also providing a statistical and intuitive method to present the data. The KDE plots, hence, can characterize the processing pattern by describing multivariate distribution [36,37]. The kernel density estimate of f at the point x is given by

$$\hat{f}(x, x_i; t) = \frac{1}{n}\sum_{i=1}^{n} K(x, x_i; t) \tag{12}$$

where

$$K(x, x_i; t) = \frac{1}{\sqrt{2\pi t}} e^{-(x-x_i)^2/2t} \tag{13}$$

is a Gaussian kernel that it assumes that the deviation of observations should be derived from the stochastic processes like jitter in measuring. $\sqrt{t}$ is referred to as the bandwidth, which influences the smoothness of the density reconstruction.

As stated in Table 2, when using a duckbill nozzle with a slit length of 11 cm, the diameter D is on average 9.6% larger than the diameter with the nozzle of 7 cm slit length (40.317–44.179 μm). Whereas, the depth did not change as much since the depth was only 2.9% higher with 11 cm slit length nozzle (14.184–14.599 μm). The range of diameters is also much larger (Std_D of 1.45 μm with 11 cm slit length and Std_D of 2.036 μm with 7 cm slit length nozzle) than the range of depths (identical Std_h of ˜0.9 μm). This can also be observed in Figs. 15 and 16, where the "outliers" in diameter are obviously more present than those in depth. The microcavities' depth is definitely smaller than its diameter, therefore, more sensitive to measurement errors. Each of the measurements should be centered around the observation with a fixed

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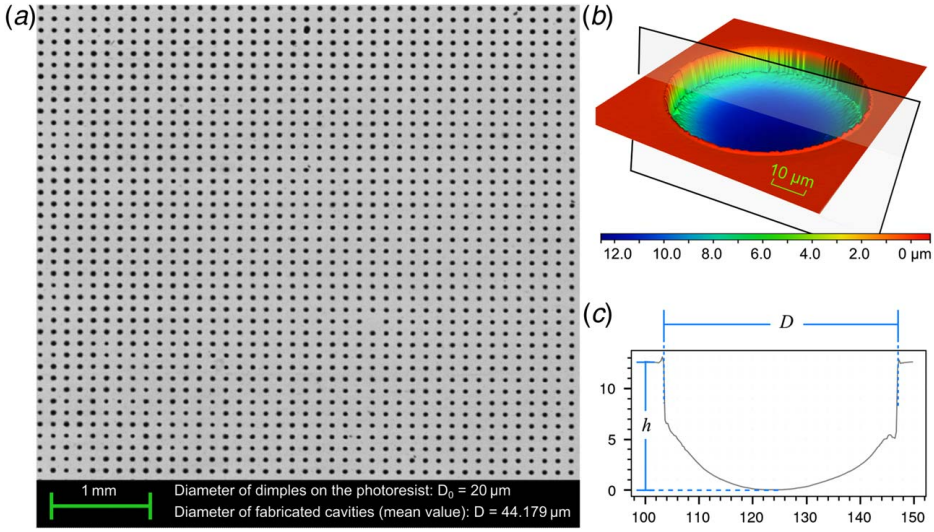

Fig. 13 Microcavities processed by MEJM with a duckbill nozzle: (*a*) microcavities fabricated by MEJM with duckbill nozzle, (*b*) 3D profile of microcavities, and (*c*) cross-sectional profile of microcavities

range. Therefore, the between-group difference in depth is little, as the measurement error is the primary contributor. The experimental results denote that the microcavities will have a higher *EF* when using a duckbill nozzle with a shorter slit length, but with less processing consistency. When using a slit length of 7 cm, the Std_{EF} is almost twice as high as that of the nozzle with a slit length of 11 cm (0.197–0.104).

In order to understand the experimental results, a physical model is built as shown in Fig. 17. The nozzle slit length ranged from 5 cm to 15 cm with a step of 0.5 cm. In the present experimental study, the machining area comprised several 35 × 35 cavity arrays to fit the lithography working space of a 3-in diameter circle, hence, the machining region in this case was simplified to 5 cm in the physical model.

As illustrated in Fig. 18, the curves of electric current density over the workpiece were presented in a shape of a small hill, the higher values are situated below the duckbill nozzle as can also be inferred from the electric potential distribution. These curves are "widening" as the slit length increases. While the machining areas were fixed, it can be assumed that a flat curve of current density in the machining area could be achieved with a long enough nozzle slit length. This could also be observed in Fig. 19. Therefore, the increasing trend of Std_D with a smaller slit length in Table 2 can be confirmed. The variation of electric current

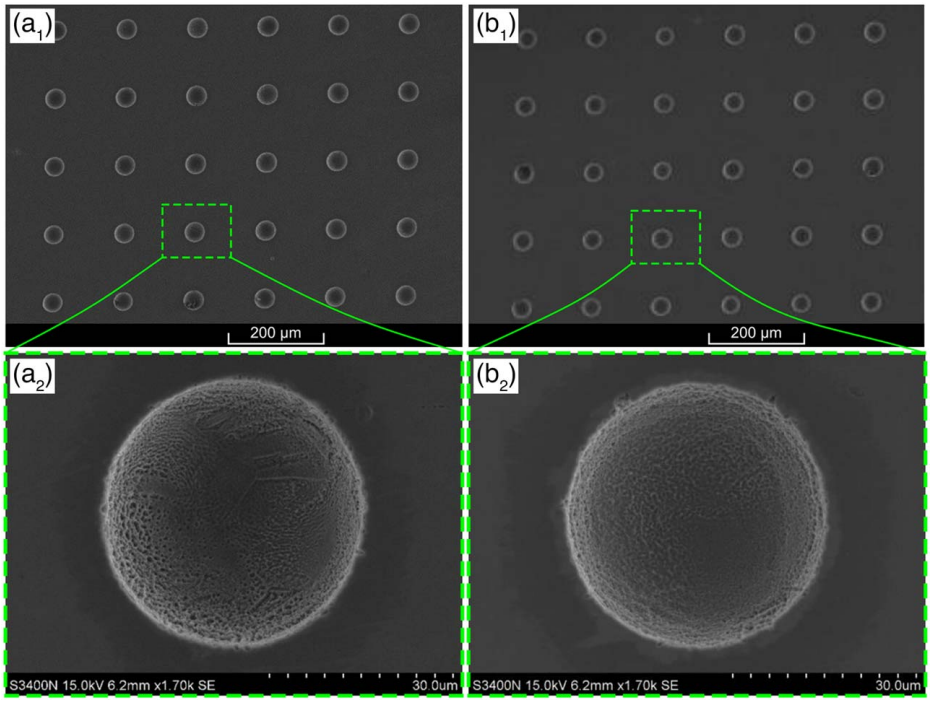

Fig. 14 Microcavities processed by MEJM with a duckbill nozzle of different slit lengths: (a_1, a_2) slit length = 11 cm and (b_1, b_2) slit length = 7 cm

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Table 2 Experimental results at different duckbill slit lengths

	Diameter D (μm)		Depth h (μm)		AR		EF	
Nozzle slit length (cm)	11	7	11	7	11	7	11	7
mean	44.179	40.317	14.599	14.184	0.331	0.353	1.212	1.413
std	1.450	2.036	0.922	0.896	0.023	0.033	0.104	0.197
min	41.572	35.611	12.269	12.330	0.267	0.288	0.944	1.084
25%	43.243	38.856	13.812	13.602	0.314	0.333	1.142	1.288
50%	44.167	40.583	14.587	14.051	0.333	0.351	1.230	1.387
75%	45.379	41.799	15.191	14.706	0.348	0.371	1.267	1.534
max	46.711	44.786	16.640	16.038	0.373	0.433	1.397	1.975

Note: $N = 50$.

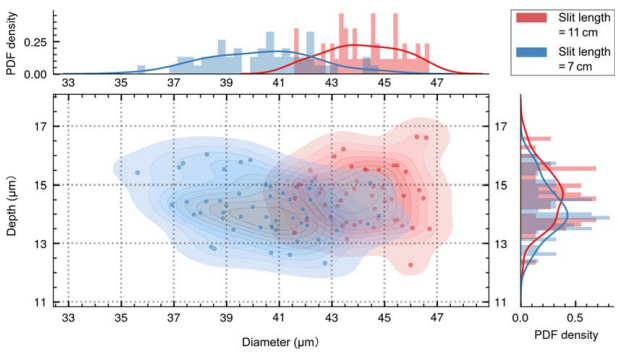

Fig. 15 Diameter, depth, and corresponding PDF density of microcavities

density distribution in the machining area induces the variation of machined cavities, resulting in the observed decentralization of diameter D in Fig. 15.

On the other hand, the peak value of the electric current density on the workpiece was higher with a longer nozzle slit length. A higher current density value can cause more material removal rate and consequently a deeper and larger cavity, this also explains that in Table 2, when using a duckbill nozzle with a slit length of 11 cm, the maximum value of diameter D_{max} is larger than the D_{max} with the nozzle of 7 cm slit length (46.711–44.786 μm), so does the maximum value of depth h_{max}(16.640–16.038 μm). The differences between the diameters of the machined cavity for each nozzle type and differences in the machined cavity dimensions for different nozzles can both be prompted by the current density distribution.

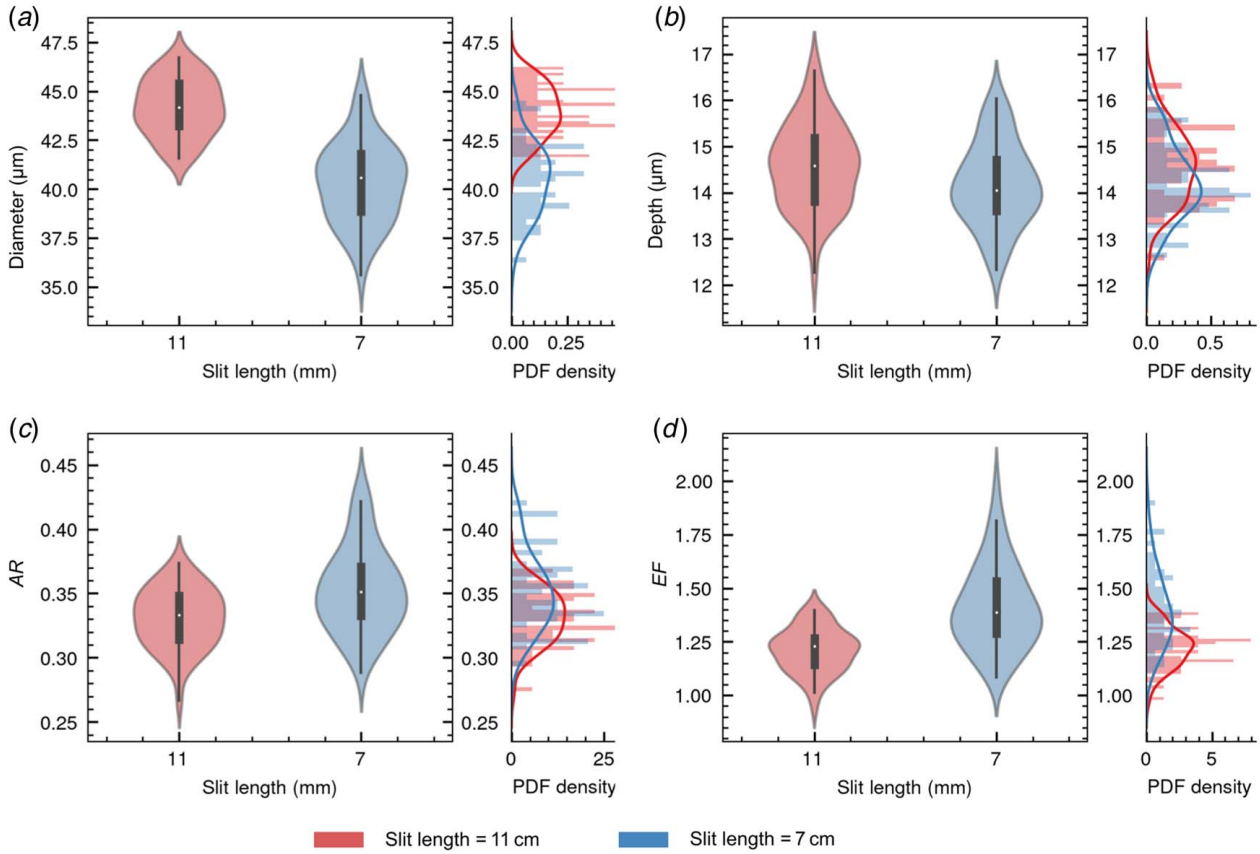

Fig. 16 Diameter, depth, *EF* value, *AR*, and corresponding PDF density of microcavities: (*a*) the diameter of processed microcavities, (*b*) the depth of processed microcavities, (*c*) the *AR* of processed microcavities, and (*d*) the *EF* of processed microcavities

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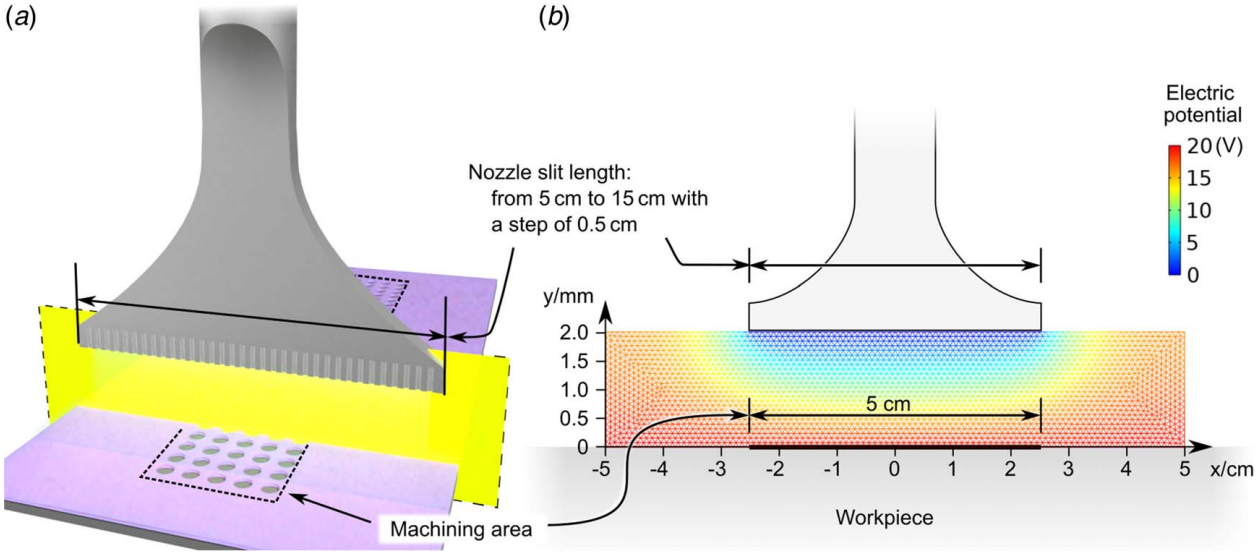

Fig. 17 Schematic view of the physical model for analyzing the relationship between the nozzle slit length and machining area: (*a*) 3D schematic view of machining area and slit length of duckbill nozzle. The dashed-border box denotes the cross-sectional area where the electric field study carried out and (*b*) electric potential distribution and corresponding mesh of physical model.

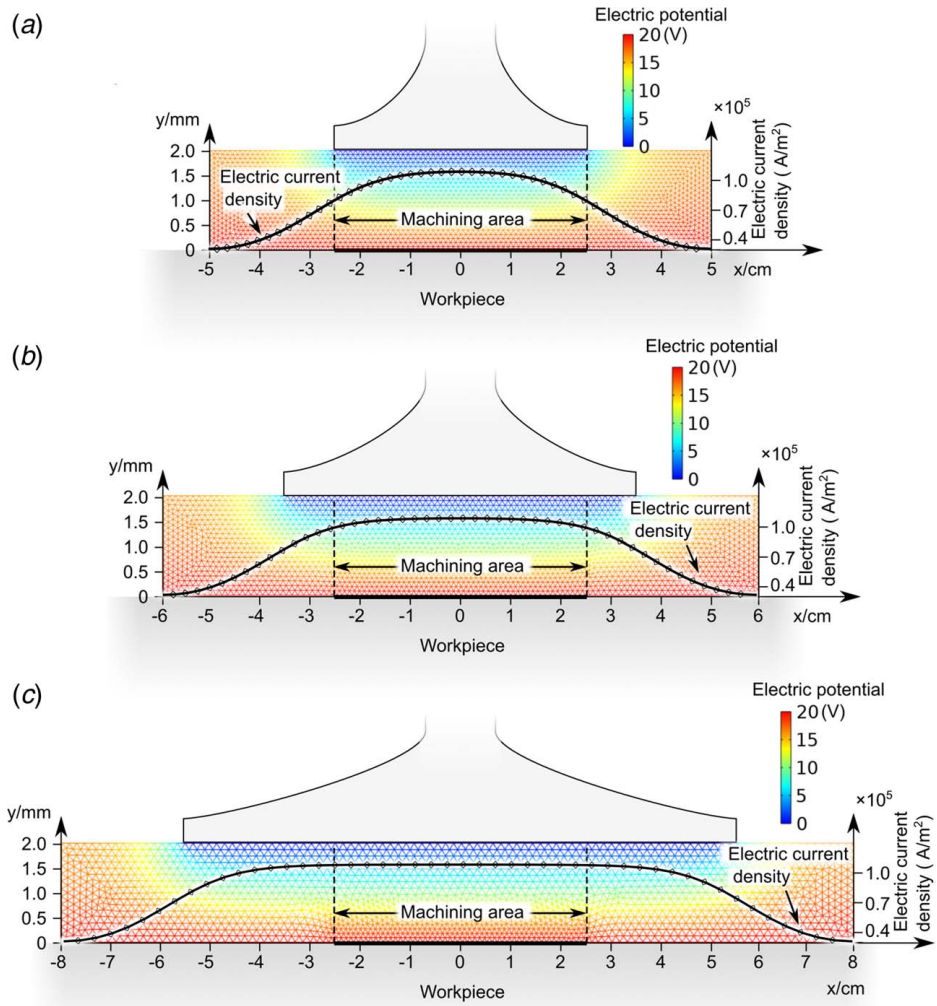

Fig. 18 Electric current density on the workpiece with different nozzle slit lengths: (*a*) electric current density with a slit length of 5 cm, (*b*) electric current density with a slit length of 7 cm, and (*c*) electric current density with a slit length of 11 cm

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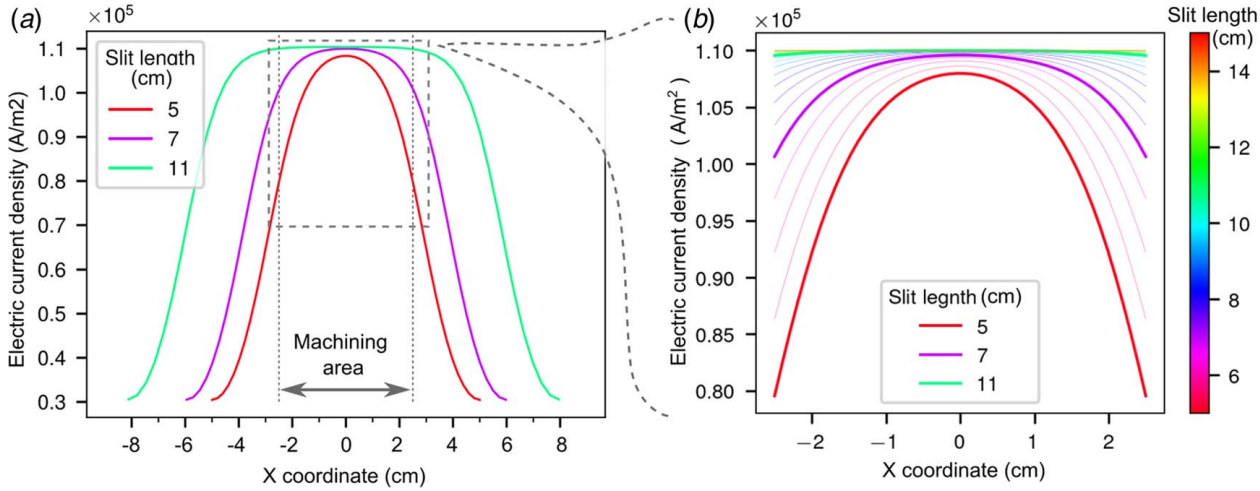

Fig. 19 Electric current density on the workpiece with different nozzle slit lengths: (*a*) electric current density on the workpiece and (*b*) electric current density on the machining area with different slit lengths

As shown in Figs. 19(*a*) and 19(*b*), the peak values of current density, i.e., the amplitude, are different with different slit lengths. While in practice the nozzle type is usually not changing, therefore, only the within-group deviations of current density should be outlined, and the between-group differences in amplitude should also be eliminated. The coefficient of variation (*CV*) given in Eq. (14) is independent of the range or mean value of observations, therefore, often used as a dimensionless metric for between-group data [38]

$$CV = \frac{\sigma}{\mu} \times 100\% \tag{14}$$

where σ is the standard deviation and μ is the mean value of observations. Here, the current density curves in Fig.19(*b*) were interpreted as *CV* values that are free from the data range. In order to present a general insight into the relationship between the machining area and nozzle slit length, the slit length was converted to a percentage of the machining area, i.e., the dimensional ratio R_d:

$$R_d = \frac{\text{Slit length}}{\text{Machining area}} \tag{15}$$

The interpreted results are shown in Fig. 20. As mentioned above, the machining area is 5 cm, thus a slit length of 7 cm is equal to the dimensional ratio $R_d = 120\%$, and a slit length of 11 cm is equal to the dimension ratio $R_d = 220\%$, the corresponding *CV* is 2.273–0.107% which was in good agreement with the experimental results in Table 2. In addition, two thresholds of the dimensional ratio R_d are presented for the processing consistency:

(1) The performance threshold

The performance threshold is determined by the *CV* which can hardly make any statistical difference and is given in Algorithm 1 (Procedure: P-Threshold). In the present study, $CV = 0.0463\%$ in which $R_d = 300\%$ is taken as CV_t to illustrate the following efficient threshold (the red parallel dash lines in Fig. 20). Indeed, when the $R_d = 200\%$, the corresponding *CV* is 0.234%, which is already small enough to be assigned as the performance threshold to produce any statistical difference.

(2) The efficient threshold

Theoretically, as the slit length approaches infinity, the *CV* will approach 0% and then will eliminate any processing inconsistency caused by the electric field. However, this is apparently impossible. Therefore, the efficient threshold is given (green spot in Fig. 20) for making a cost-performance tradeoff. When the dimensional ratio R_d is lower than the efficient threshold, the slope of *CV* is lower than the slope of the entire data sets, which means the *CV* will drop faster than the increasing R_d. And vice versa when the R_d is larger than the efficient threshold. As illustrated by the red parallel dash lines in Fig. 20, the efficient threshold can be determined by Algorithm 2 (Procedure: E-Threshold).

Algorithm 1 Performance threshold

```
CV(i) is the i-th value of CV, R_d(i) is the i-th value of R_d
CV_t is the minimum value of CV which have the statistical difference
I is the total number of data points
procedure P-THRESHOLD (i, CV_t)
  for i ≤ I do
    if CV(i) == CV_t then
      t ← i        ▷Assign the value of i to the value of t
    end if
  end for
  if t ∄ then
    t ← I
  end if
  Performance threshold ← R_d(t)
end procedure
```

Algorithm 2 Efficient threshold

```
S(i) the i-th value of the slope
procedure E-THRESHOLD (t)
  for i ≤ t do
    S(i) ← (CV(i)−CV(i+1)) / (R_d(i)−R_d(i+1))
    if S(i) == S(t) then
      Efficient threshold ← R_d(i)
    end if
  end for
end procedure
```

The machining variation in batch fabrication is one of the most critical interests for all the mask-based ECM technologies, including TMEMM, MEJM, etc., to enable industrial applications. In the present study, initially, the duckbill nozzle was intended to make MEJM more capable of batch microfabrication by covering more processing areas. Surprisingly, the use of a duckbill nozzle

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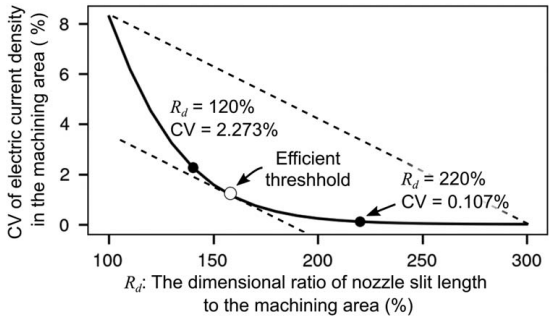

Fig. 20 Dimensionless interpretation of Fig. 19 and the efficient threshold

is offering an extra advantage in processing consistency. By interpreting the processing parameter in the present study to the dimensionless value, general instructions for all the mask-based ECM processes can be collected. In principle, the cross-sectional width of the tool should be at least two times larger, i.e., the aforementioned performance threshold, than the cross-sectional width of the workpiece to wipe out any observable deviation of current density distribution in the machining area.

3.2 Investigation of Underetching by the Virtual Gap and Discrete Model. The presence of the virtual gap in the multiphysical model makes it possible to simulate the underetching in the mask-based ECM process [21]. Discretizing a physical model into independent subdomains according to their size scale ensures a more precise description, as shown in Fig. 21, for the underetching as well as the processing feature evolution. Since the mesh size close to the electrochemical reaction interface can be refined into a nanoscale in the present study, the mesh on the reaction interface has a minimum size of 10 nm, which provides an unprecedented accuracy with reasonable computational costs for underetching analysis.

Experimental studies were also carried out along with the numerical model to verify the proposed model. The experimental parameters are the same as in Sec. 3.1, except for two different diameters of dimples on the photoresist, $D_0 = 5\ \mu$m and $D_0 = 20\ \mu$m, that were used to test the generalization ability of the proposed method. As

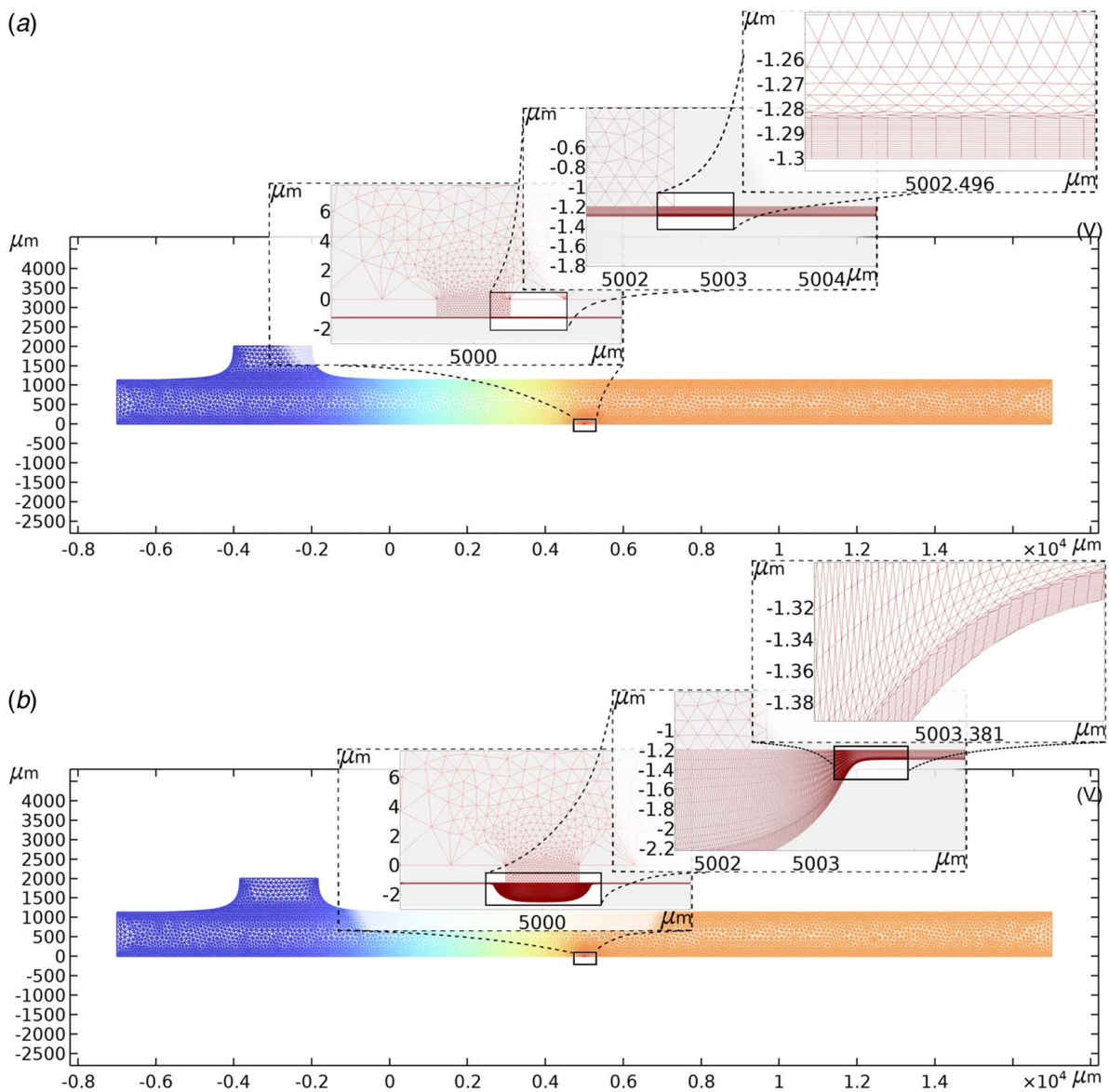

Fig. 21 Electric potential distribution and mesh at different time-steps: (*a*) electric potential distribution and mesh at $t = 0$ s and (*b*) electric potential distribution and mesh at $t = 0.8$ s

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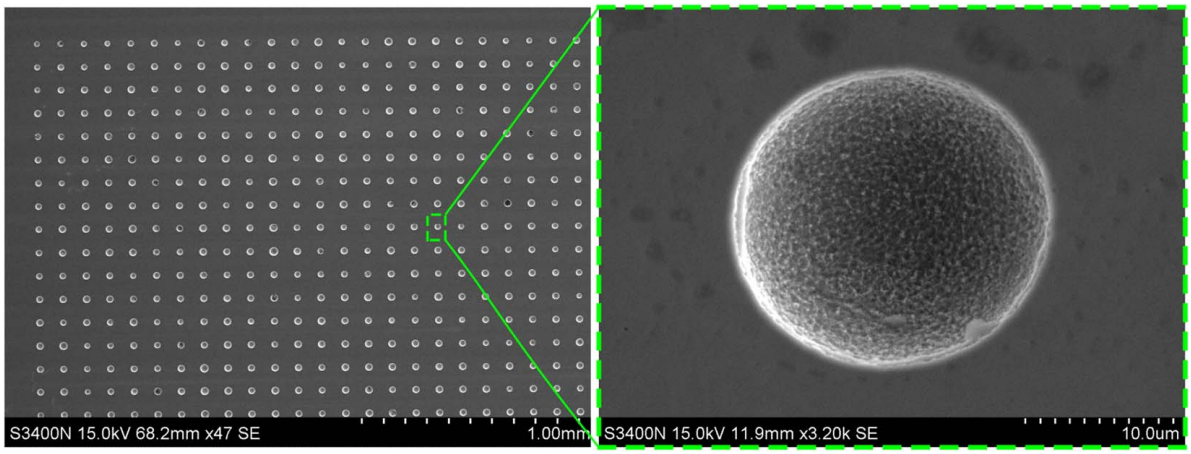

Fig. 22 Microcavities fabricated by MEJM, the corresponding diameter of microdimples on the photoresist is 5 μm

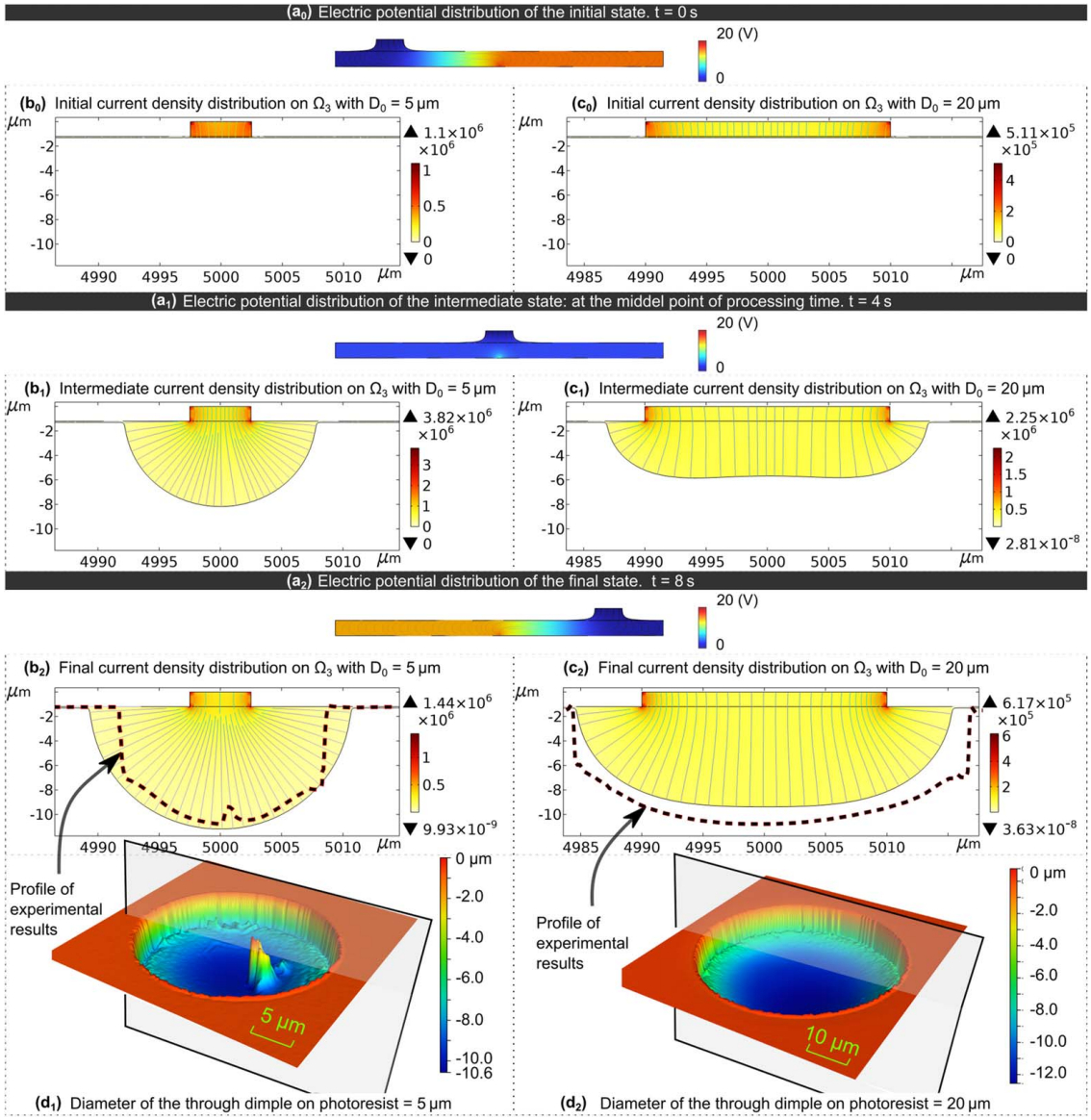

Fig. 23 Simulation and experimental results for fabrication of microcavities with different diameters of dimples on photoresist by MEJM: (a_0–a_2) electric potential distribution at different time/nozzle positions, (b_0–b_2) electric current density distribution in the domain Ω_3 with $D_0 = 5\ \mu$m; (c_0–c_2) electric current density distribution in the domain Ω_3 with $D_0 = 20\ \mu$m, and (d_1, d_2) experimental results with corresponding D_0

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shown in Fig. 22, microcavities were processed by MEJM, the corresponding diameter of microdimples on the workpiece is 5 μm.

As stated in Fig. 23:

The overall electric potential distribution with different processing times is shown in Fig. 23(a_0–a_2).

At the corresponding time, the electric current density distribution in deformation domain Ω_3 (q.v. Fig. 4) is shown in Figs. 23(b_0–b_2) and (c_0–c_2) with different D_0.

The corresponding cross-sectional profiles of the experimental results are shown in Fig. 23(d_1, d_2).

From a macroperspective (Fig. 23(a_0–a_2)), at the beginning of the machining process ($t = 0$ s) and at the end of the machining process ($t = 8$ s), the nozzle is quite far away from the exposed cavity (8 mm), therefore, the electric potential difference should be quite small in the Ω_3 domain. A drastic change in the electric potential difference or the active dissolution process only happens when the nozzle (cathode) is directly above the exposed cavity in the deformation domain Ω_3.

From a microperspective, the electric current density distribution in Ω_3 showed that electric current density peaks are always present at the edge corner of the photoresist on the workpiece, which corresponds to the conclusions of Sec. 2.2.2 on virtual gap size refinement.

The peak values of current density in Fig. 23(b_2/c_2) are higher than that in Fig. 23(b_0/c_0) which may seem counterintuitive. The present model is symmetrically designed, the deformation domain Ω_3 is at the center of symmetry, so the distance from Ω_3 to the nozzle is exactly the same whether at the beginning or at the end of the process. The reaction interface, on the other hand, moves downward during the machining, resulting in a definitely larger IEG at the end of the process. Therefore, a smaller peak electric current density in Fig. 23(b_2/c_2) should be expected at the process end.

However, the reverse pattern was observed. It is well known that the electric field is affected by geometry, more precisely, the curvature. In our case, the underetching will make the photoresist "sharper" by "excavating" the materials below. This "sharpening" effect not only canceled the negative current density offsets activated by the increasing IEG but also even surpasses them. The geometric shape of the mask-workpiece combination may play a more important role in the evolution of microsurface structures prepared by a mask-based technology. As shown in Fig. 23(d_1, d_2) and the cross-sectional profiles in Fig. 23(d_1, d_2), there are obviously "cliffs" on the side wall of machined cavities, these cliff-like artifacts are from the ECM process and not from the measurement technique. When compared with simulation results, the depth of "cliff" is almost the same length of underetching. One has to wonder whether there is a connection between the underetching and the "cliffs" on the side wall of machined cavities.

While inspecting mask-based ECM process investigations by other researches, much to our surprise, the "cliffs" were widespread across fabrication of cavities [11,17,21,28], channels [36], etc. Therefore, it is reasonable to suppose that the mask, which in most cases is of excellent elastic resilience (Shipley 1818 in this work), was bending inward to the bottom of the cavity because of the underetching and the electrolyte pressure during the processing time, as illustrated in Fig. 24. Since the bending mask would protect the side wall of machined cavities, consequently the bottom side would be exposed to a higher intensity of electrochemical attack, cavities with smaller underetchings and deeper depths were expected. The same hypothesis can also be extended to real 3D machining cases where the mask may be subjected to warping, stretching, or tearing as well depending on the mechanical properties of mask. In these cases as well, the side wall of cavities is dissolved less in comparison to the bottom side. Thus, this hypothesis also explains that despite the electrochemical reactions are isotropic reaction in nature, the *EF* value as shown in Table 2, is higher than 1. If this assumption holds true, theoretical and computational analysis for mask-based ECM should address more attention on the soft material in the solid–fluid multiphysical coupling, the "virtual gap and discrete modeling" strategy offers in that context significant structure-aided framework for further analysis. Future work will involve development of an on machine microscopic observation technique to observe mask bending/warping to support the hypothesis.

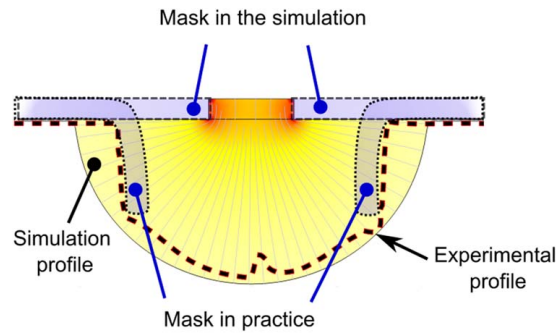

Fig. 24 Hypothesis of the bending mask

4 Conclusion

The consistency of feature dimensions in batch fabrication of microstructures always receives extensive attention for industrial applications. In this study, a method of mask electrolyte jet machining with a duckbill nozzle was proposed to boost the machining capacity and the processing consistency. In the meantime, multiscale modeling coupled with multiphysics is an emerging scientific discipline as well as the wide-spread application of macro–microstructures fabricated by ECM. This study proposed a high-performance numerical analysis framework for this type of ECM technology.

The conclusions of the present work can be summarized as follows:

(1) Discretizing the model into subdomains with independent mesh size can reduce the computational cost. Finer mesh for the regions of significant interest and coarser mesh for the regions away from the processing zone which is of less interest.

It has been shown that on the fine–coarse interface the amount of nodes on the coarse mesh side should at least be 20% amount of nodes on the fine mesh side.

(2) Introducing the concept of virtual gap can illustrate the underetching for all the mask-based ECM processes. Besides it also represents altering the IEG and causes a deviation in the current density distribution on the reaction interface.

When the virtual gap is less than 100 nm, the current density deviation is mostly prompted by offsetting the sharp corner. Therefore, the virtual gap of less than 100 nm is sufficient to capture the effects of geometrical influence without unnecessarily increasing the IEG.

(3) With a larger slit length, the duckbill nozzle can not only cover more processing area but also grants a more uniform current density distribution over the entire machining area, leading to fabrication of microstructures with better consistency. For instance, this work has successfully fabricated several microcavities arrays with a diameter range of 40–46 μm and a depth range of 12–16 μm depending on the applied parameters. The processing consistency was boosted when using a larger slit length. This work enabled the definition of two thresholds:

Performance threshold: The slit length of the duckbill nozzle should be at least two times larger than that of the machining area to wipe out any observable deviation on the current density distribution in the machining area.

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Efficient threshold: The slit length cannot be arbitrarily large. Therefore, the efficient threshold is given for the cost-performance tradeoff. When the dimensional ratio R_d is lower than the efficient threshold, the process consistency will increase faster. And vice versa when the R_d is larger than the efficient threshold.

(4) As the mask in the present study is of excellent elastic resilience, there are "cliffs" on the side wall of processed cavities. These cliffs may be produced by the mask bending, which is provoked by the underetching and electrolyte fluid pressure.

The "virtual gap and discrete modeling" strategy may also be extended for large-scale translational movement analysis, such as Jet-ECM with a moving jet nozzle. The efficient threshold and performance threshold can provide instructions to design the machine tool system for all the mask-based ECM processes, such as MEJM and TMEMM. Overall, this work provides not only a next step toward using MEJM technology for deterministic and efficient fabrication of microstructures such as cavities/grooves on large workpieces but also introduces an implementation of the machine tool-workpiece system which can improve the processing consistency, and provides an efficient high-performance numerical simulation framework. Both the implementation and the computational framework can be easily implemented and easily extended for all the mask-based ECM processes.

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Conflict of Interest

There are no conflicts of interest.

Data Availability Statement

The datasets generated and supporting the findings of this article are obtainable from the corresponding author upon reasonable request.

Appendix

The photoresist mask was prepared by the following procedures:

Substrate preparation: The workpiece (in this case, stainless steel) was polished to a mirror-like surface (surface roughness $\leq 8\,\mu$m) and then ultrasonically cleaned in alcohol and acetone as the substrate.

Photoresist coating: The photoresist (in this case, Shipley 1818) was deposited on the substrates by spin-coating (5000–10,000 rpm) to produce a thin photoresist film (1.2 μm).

Soft bake: After the photoresist film was prepared, it was baked on a hotplate at 115 °C for 60 s to remove solvents and solidify the film.

Exposure: The photoresist film was patterned by using an ultraviolet exposure (150 mJ/cm^2) and a chrome mask.

Development: The double puddle spray (purchased from Shipley Company Inc.) was used to develop the pattern on the workpiece.

Hard bake: After the Development, a hard bake at 90 °C for 300 s was performed to solidify the patterned photoresist on the workpiece.

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